MAST20026 REAL ANALYSIS 7B

One-Sided Limits, Limits to Infinity and Boundedness

Learning Incomes

- Familiarity with the $\epsilon - \delta$ definition of one-sided limits.
- Familiarity with the definition of the notation $\lim_{x\to\infty} f(x)$ and $\lim_{n\to\infty} f_n$.

1. **One-Sided Limits**
   Let
   $$f(x) = \begin{cases} 2x & x > 1 \\ -2x & x < 1 \\ 0 & x = 1 \end{cases}$$
   (a) Using an $\epsilon - \delta$ argument, prove $\lim_{x\to 1^-} f(x) = -2$.

   (b) Using an $\epsilon - \delta$ argument, prove $\lim_{x\to 1^+} f(x) = 2$.

   (c) By citing a theorem from the lecture notes, prove the function diverges at $x = 1$.

2. **Sequences and Functions Part I**
   Let $f(x) = \frac{4x}{x+3}$. For every $n \in \mathbb{N}^+$, let $f_n = f(n)$.

   (a) Using the definition of convergence as x goes to ∞, prove
   $$\lim_{x\to\infty} f(x) = 4$$
   (b) In the previous part, you have shown that for every $\epsilon > 0$, there exists $M \in \mathbb{R}$ so that $|f(x) - 4| < \epsilon$ whenever $x > M$. Using this fact, prove $\lim_{x\to\infty} f_n = 4$

3. **Sequences and Functions Part II**
   Let $f : \mathbb{R} \to \mathbb{R}$ be a function and define a sequence (f_n) so that $f_n = f(n)$ for all $n \in \mathbb{N}^+$.

   (a) Let $L \in \mathbb{R}$. Using the definition of convergence for sequences and functions prove that if $\lim_{x\to\infty} f(x) = L$, then $\lim_{n\to\infty} f_n = L$.

   (b) Notice that the convergence of (f_n) depends only on the behaviour of $f(x)$ for $x \in \mathbb{N}^+$ Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ so that f_n converges as x goes to ∞, but $f(x)$ diverges as x goes to ∞.

4. **Limits and Boundedness**
   In our study of sequences, we proved that every convergent sequence is bounded. Here we consider an analogue of this result for functions approaching a limit point. We prove that if $f(x)$ converges as x approaches a, then $f(x)$ is *bounded* when x is *sufficiently close* to a.

   (a) Let E be a set, let a be a limit point of E and let $\delta > 0$. Let $N_0(a, \delta)$ denote the following set
   $$N_0(a, \delta) = \{x \in \mathbb{R} \mid 0 < |x - a| < \delta\}$$
   Using the definition of limit point, explain how you know $N_0(a, \delta)$ is always non-empty.

   (b) Let $L \in \mathbb{R}$, $f : \mathbb{R} \to \mathbb{R}$ and $a \in \mathbb{R}$. Prove that if $\lim_{x\to a} f(x) = L$, then there exists δ so that the set
   $$\{f(x) \mid x \in N_0(a, \delta)\}$$
   is bounded above and below in $\mathbb{R}$. (Hint: What can you conclude when $\epsilon = 1$?)

   (c) State the contrapositive of the implication in the previous part.

   (d) Prove $\lim_{x\to 0} \frac{1}{x^2}$ diverges using the statement in previous part.

5. CHALLENGE **Sequences and Functions Part III**
Let $f : E \to \mathbb{R}$ be a function, and let a be a limit point of E.

(a) Suppose that $\lim_{x \to a} f(x) = L$. Show that if (a_n) is a sequence such that $a_n \to a$ and $a_n \in E \setminus \{a\}$, then $\lim_{n \to \infty} f(a_n) = L$.

(b) Suppose now that for all sequences (a_n) converging to a and contained in $E \setminus \{a\}$, $\lim_{n \to \infty} f(a_n) = L$. Prove that $\lim_{x \to a} f(x) = L$.

Hint: Try a proof by contraposition: If $f(x)$ does not converge to L as $x \to a$, can you produce a sequence (a_n) in $E \setminus \{a\}$ such that $a_n \to a$ but $(f(a_n))$ does not converge to L?